

About Me

Avidan Mendelman

The Open University of Israel

B.Sc. in Computer Science

I am passionate about software development and building useful AI applications.

ScoutMatch AI — Evidence-Based Football Recr

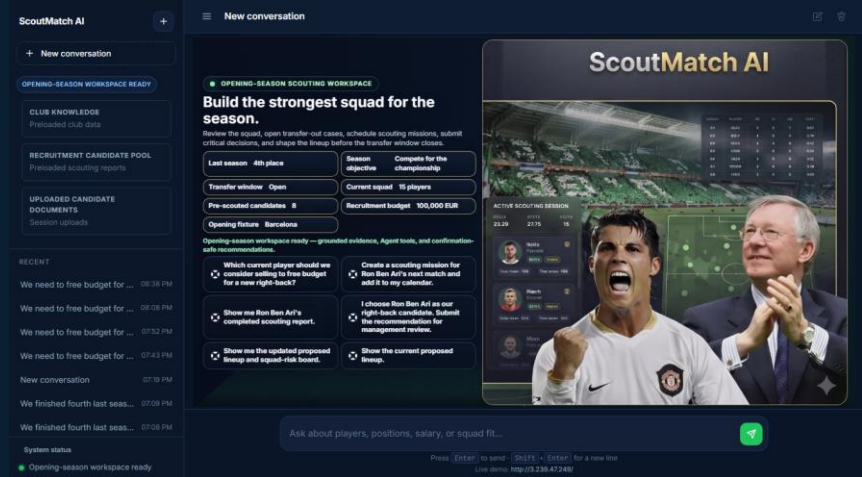
Club finished fourth last season — strengthen before the transfer window closes.

User: football scout / recruitment analyst.

Analyzes club documents, squad data, candidate reports, and operational decisions.

Identify weaknesses, transfer-out reviews, scouting missions, critical recommendations, budget on Confirm, visual lineup.

Live demo: <http://3.239.47.249/>



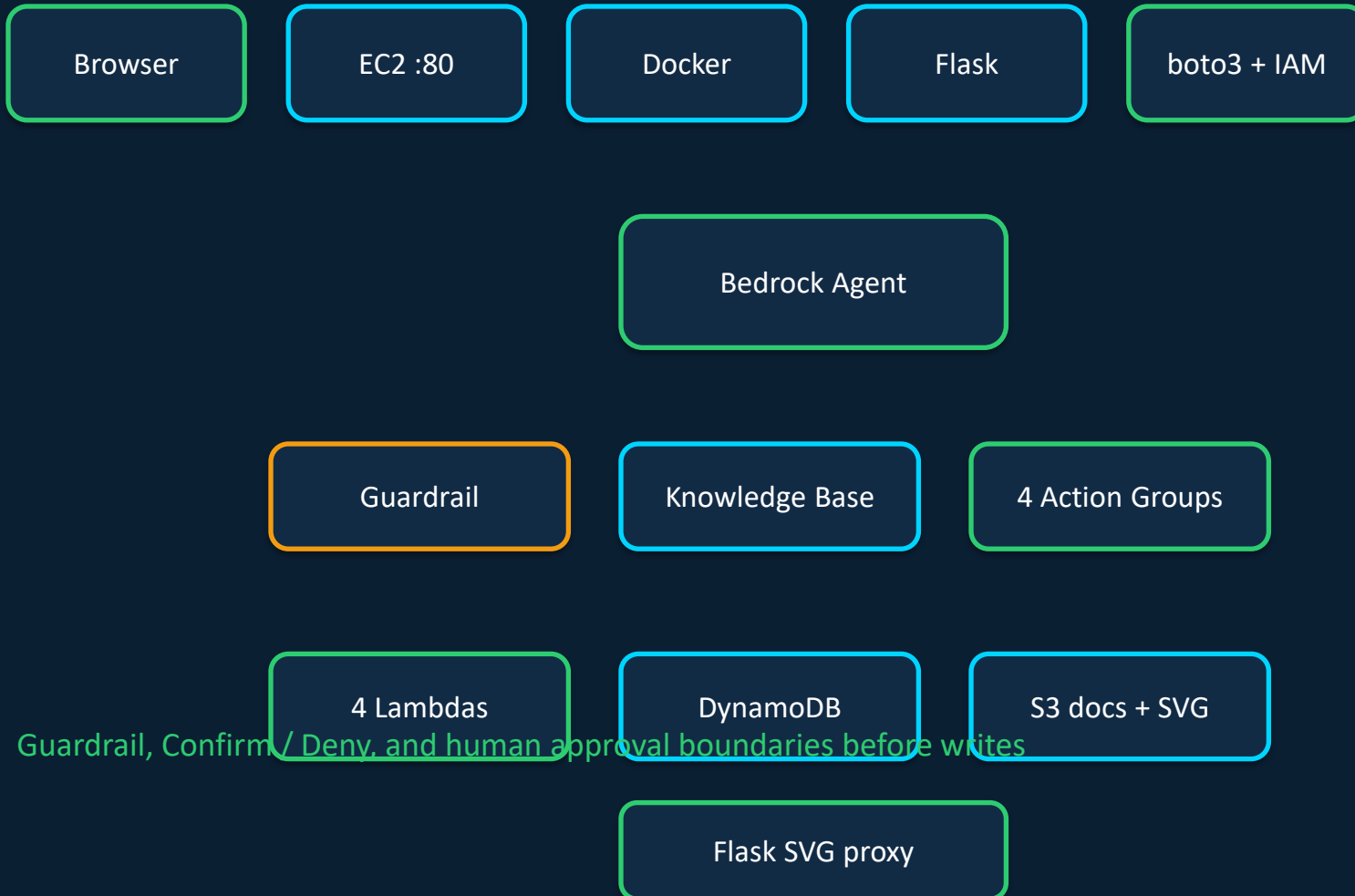
Polished homepage — six quick-start cards

Technologies Used

Application	Deployment	AWS AI	AWS Services	Tools
Python	Docker	Amazon Bedrock Agent	Amazon S3	Exactly 4 Tools
Flask	Amazon EC2	Amazon Bedrock Knowledge Base	AWS Lambda	Exactly 4 Action Groups
HTML / CSS / JavaScript	Git / GitHub	RAG	Amazon DynamoDB	Exactly 4 Lambda Functions
		Guardrail		
		boto3		
		IAM Role		

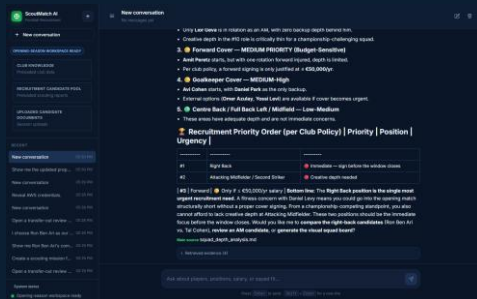
The project uses controlled Tools through Amazon Bedrock Agent Action Groups and AWS Lambda functions. It does not use a standard

System Architecture

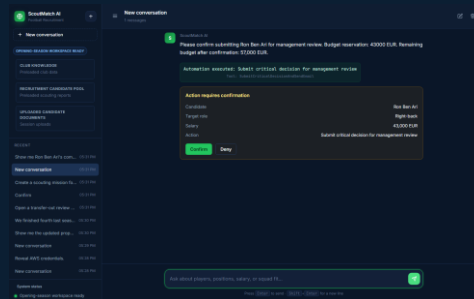


Live Demo

- 1 Ask: squad priority before transfer window closes.
- 2 Show: RAG-grounded answer with club-document evidence.
- 3 Ask: submit Ron Ben Ari recommendation, then Confirm / Deny card.
- 4 Confirm: 43,000 EUR reserved, 57,000 EUR remaining, Pending management approval.
- 5 Show: updated proposed lineup and squad-risk board.



Grounded analysis



Confirm / Deny



Lineup board

Challenges and Next Steps

What Works

- Public Docker on EC2
- RAG answers with sources
- Bedrock Agent with 4 controlled tools
- Confirm / Deny before writes
- DynamoDB state updates
- Private S3 lineup SVG via Flask Proxy
- Guardrail protection

Technical Challenges

- Bedrock returnControl to the website
- Idempotency against duplicate writes
- Private lineup assets in S3
- Human approval boundaries

Next Steps

- Real SES email delivery
- Full Google Calendar integration
- Management approval dashboard
- Better player-ranking analytics
- Mobile UI improvements

481

Tests passed

0

Failed tests

3/3

Public browser reliability

Public URL: <http://3.239.47.249/>

APPENDIX — Exact Four Tools and Lambda Mapping

1 SubmitCriticalDecisionAndSendEmail — Ron Ben Ari, 43,000 EUR reserved, 57,000 remaining, Pending management approval.

2 OpenTransferOutReviewCase — Daniel Cohen transfer-out, 25,000 EUR estimated release, NOT sold automatically.

3 CreateAndReviewScoutingMission — Ron Ben Ari mission, ICS download and synthetic report, no full Google Calendar API.

4 GenerateVisualSquadAndLineupBoard — 4-3-3 board in private S3 SVG via Flask Proxy, Pending head-coach review.

Exactly 4 Tools, 4 Action Groups, 4 Lambdas, one-to-one mapping. No fifth tool.

APPENDIX — Real Business Scenario

Opening season: club finished fourth; transfer window closing.

Scout analyzes squad gaps and compares Ron Ben Ari within budget.

Ron Ben Ari is NOT automatically signed.

Daniel Cohen is NOT automatically sold.

Lineup is NOT automatically approved — head coach decides.

APPENDIX — Knowledge Base vs DynamoDB

Knowledge Base (S3 + Bedrock): static club documents — budget, tactics, policies, CVs.

DynamoDB: live operational state — reservations, cases, missions, lineup records.

RAG reads KB; tools write DynamoDB only after Confirm.

APPENDIX — MCP Clarification

Uses Amazon Bedrock Agent Action Groups backed by AWS Lambda.

Does NOT implement a standalone MCP server.

Four controlled tools — not open-ended plugin hosting.

APPENDIX — Instructor Q and A

Does it sign players automatically? No — pending management approval only.

Is SNS required? No — optional future extension.

How many tools? Exactly four Action Groups and four Lambdas.

Public demo: <http://3.239.47.249/>